

BHAVANI AMBATI *DevOps Engineer*

bhavani.a562@gmail.com

+1 9375160521

USA

Linkedin

GitHub

Profile

DevOps Engineer with around **5 years** of experience designing and operating production-grade cloud infrastructure and GitOps-driven CI/CD platforms. Expert in **AWS, Terraform, Kubernetes (EKS), Jenkins, Docker, Helm, Python**, and **observability** stacks (Prometheus/Grafana/ELK). Proven track record of accelerating deployments, driving cost optimizations, and implementing DevSecOps at scale. Strong collaborator in Agile teams with experience in cloud migrations, governance, and SRE practices.

Professional Experience

DevOps Engineer, Capital One

Oct 2024 – Present | Remote, USA

- **Designed and implemented GitOps CI/CD pipelines** with **Jenkins, GitHub Actions, ArgoCD, and Spinnaker**, automating build/test/deploy for **50+ microservices** and reducing release cycle time from **2 days to <4 hours**.
- **Automated multi-account AWS infrastructure** using **Terraform, CloudFormation, and Ansible**, enforcing policy-as-code and audit-ready provisioning across environments.
- **Containerized and migrated services to Amazon EKS**, implementing **Helm-based blue/green and canary rollouts**, auto-scaling, and RBAC controls to achieve **99.99% availability**.
- **Built Python/Bash/SQL automation** for deployment validation, EKS node rotation, snapshot lifecycle, and security audits, cutting manual runbook tasks by **~70%**.
- **Implemented observability and distributed tracing** with **Prometheus, Grafana, OpenTelemetry, ELK, and CloudWatch**, enabling proactive alerts and reducing MTTI/MTTR.
- **Embedded DevSecOps** into pipelines (SonarQube, Snyk, Checkov, Vault, Aqua), enforcing vulnerability gating and achieving 100% scheduled compliance.
- **Governance & cost controls**: standardized IAM, tagging strategy, lifecycle rules and reserved-instance planning — saved **>\$25K/year**.
- Mentor/partner: collaborate with product, security, and architecture teams; mentor junior engineers and run internal brown-bag sessions.

Development Operations Engineer, Infor (India) Private Limited

Aug 2021 – Jul 2023 | Hyd, India

- Implemented and maintained **CI/CD pipelines (Jenkins, Git)** enabling **300+ production deployments** with versioning, environment segregation, and automated rollback.
- Automated provisioning using **Terraform / CloudFormation + Python / PowerShell**, reducing environment setup time by **60%** and eliminating drift.
- Deployed Docker containers to **Amazon EKS**, authored **Helm charts**, set up namespaces and autoscaling policies to ensure high availability.
- Centralized logging and alerting with **CloudWatch, Grafana, Sumo Logic**, and Lambda health checks for proactive remediation.
- Executed lift-and-shift migration of Windows/.NET stacks to AWS, modernizing to RDS-backed architectures and improving uptime by **~40%**.
- Enforced IAM least privilege, MFA, and automated backup / DR practices for business continuity.

DevOps Engineer, Tiger Analytics

Feb 2020 – May 2021 | Hyd, India

- Built CI/CD pipelines using **GitLab CI & Jenkins** to deploy data pipelines and ML services; automated model packaging and release workflows.
- Provisioned analytics environments in **AWS (EC2, S3, Lambda, EKS)** and **Azure (AKS)** using Terraform, enabling reproducible training/serving environments.
- Containerized ML pipelines (Docker) and orchestrated on Kubernetes for batch and near-real-time jobs; tuned resource configs to lower cost and latency.
- Automated environment bootstrap and job orchestration with Python/Bash; reduced manual provisioning time by **~70%**.
- Implemented monitoring for model drift and pipeline failures with Prometheus/Grafana and CloudWatch; integrated MLflow for model versioning.
- Integrated security scans (Trivy, Snyk) and static analysis to reduce vulnerabilities in container images.

Skills

Cloud Platforms

Amazon Web Services(AWS) (EC2, EKS, RDS, S3, Lambda, VPC, IAM, CloudFormation), Azure (AKS, VMs, Blob Storage)

CI/CD & DevOps Tools

Jenkins, GitHub Actions, GitLab CI/CD, ArgoCD, Spinnaker, Bitbucket Pipelines, Nexus, JFrog Artifactory

Containers & Orchestration

Docker, Kubernetes (EKS/AKS), Helm Charts, Argo Rollouts, Service Mesh, RBAC

Scripting & Automation

Python, Bash, Shell, PowerShell, SQL

Databases & Messaging

PostgreSQL, MySQL, AWS RDS, Kafka, RabbitMQ

Best Practices

GitOps, Blue-Green Deployments, Canary Releases, Immutable Infrastructure, Disaster Recovery, Auto Scaling, Cost Optimization

Practices & Methodologies

GitOps, Blue-Green Deployments, Canary Releases, Immutable Infrastructure, Auto Scaling, Disaster Recovery (DR), Cost Optimization

Infrastructure as Code & Configuration

Terraform, Ansible, CloudFormation, Helm, Kustomize, Packer, YAML, JSON, HCL

Monitoring & Observability

Prometheus, Grafana, CloudWatch, ELK Stack, Sumo Logic, Datadog, New Relic, OpenTelemetry, CloudTrail, Open search

DevSecOps & Security

SonarQube, Snyk, Checkov, Vault, Trivy, Aqua Security, OPA, IAM Policies, AWS WAF

Version Control & Collaboration

Git, GitHub, GitLab, Bitbucket, Jira, Confluence

Education

Master of Science in Computer Science, *Wright State University*

Aug 2023 – May 2025 | Dayton, USA

Projects

SpeechVerse++, *Distributed Real-Time Voice Command Processing System*

Designed and deployed **SpeechVerse++**, a cloud-native, real-time voice command processing platform built using **Ray**, **OpenAI Whisper**, and **Google FLAN-T5** on **AWS**. Engineered a **scalable distributed architecture** for low-latency transcription and natural language command interpretation across **GPU-accelerated and CPU-based EC2 instances**. Integrated **asynchronous streaming**, optimized **Word Error Rate (WER)**, and eliminated traditional latency bottlenecks by avoiding batch processing and single points of failure. Leveraged **S3**, **Lambda**, and **CloudWatch** for storage, compute triggers, and monitoring. Achieved robust system performance under production-like workloads, supporting **dynamic scaling**, **fault tolerance**, and real-time inference at scale. Delivered a **fault-tolerant, production-ready system** with measurable latency improvement, optimized GPU utilization, and **auto-scaling architecture** under heavy workloads.

Handwritten Digit Recognition Using CNN

Developed a deep learning model using **Convolutional Neural Networks (CNN)** with **TensorFlow** and **Keras** to recognize handwritten digits in real time. Trained the model on the **MNIST dataset** for **multi-class classification** (digits 0-9), applying preprocessing techniques such as **normalization** and **augmentation** using **NumPy** and **Pandas**. The system was designed to emulate **human visual perception** and achieved **high accuracy**, demonstrating effective performance for **real-time digit classification** in **computer vision** tasks.

Certificates

- AWS Certified Solutions Architect [🔗](#)